

Experiment No.	03
Title	Write a Program using JavaScript (Functions)
Course Outcome	CO1

1. Objective / Aim

To write a JavaScript program that demonstrates functions — declaration with no/one/multiple parameters, return values, default parameters, function expressions, arrow functions, and array iteration methods (forEach, map, filter, reduce).

2. System Requirements

Hardware

Processor	Intel Core i5 or equivalent
RAM	8 GB or above
Storage	256 GB SSD / HDD

Software

Operating System	Windows 11 Home
Code Editor	Visual Studio Code
Runtime	Node.js (JavaScript execution)
Terminal	VS Code Integrated Terminal

3. Architecture / Design

File Structure

folder structure

```
lab3_exp3/  
├── lab3_2/  
    └── exp3_2.js  <-- JavaScript functions program
```

Program Logic

- Function with no parameter — welcome().
- Function with one parameter — greet(name).
- Function with multiple parameters — add(a, b).
- Function with a return value — square(n).
- Function with string parameters — student(name, branch).
- Function with a default parameter — message(name = "Student").
- Function expression — multiply = function(a, b) {...}.
- Arrow function — divide = (a, b) => {...}.
- Array iteration methods — forEach(), map(), filter(), and reduce().

4. Source Code — exp3_2.js

exp3_2.js

```
// 1. Function with no parameter
function welcome() {
    console.log("Welcome to JavaScript");
}
welcome();

// 2. Function with one parameter
function greet(name) {
    console.log("Hello " + name);
}
greet("Akshaya");

// 3. Function with multiple parameters
function add(a, b) {
    console.log("Sum:", a + b);
}
add(10, 20);

// 4. Function with return value
function square(n) {
    return n * n;
}
console.log("Square:", square(5));

// 5. Function with string parameter
function student(name, branch) {
    console.log("Name:", name);
    console.log("Branch:", branch);
}
student("Akshaya", "AIML");

// 6. Function with default parameter
function message(name = "Student") {
    console.log("Hello " + name);
}
message();
message("Akshaya");
```

```
// 7. Function Expression
let multiply = function(a, b) {
    return a * b;
};
console.log("Multiplication:", multiply(5, 4));

// 8. Arrow Function
let divide = (a, b) => {
    return a / b;
};
console.log("Division:", divide(20, 5));

// 9. forEach() function
let marks = [80, 75, 90, 85];

marks.forEach(function(mark) {
    console.log("Mark:", mark);
});

// 10. map() function
let numbers = [1, 2, 3, 4, 5];

let double = numbers.map(function(n) {
    return n * 2;
});

console.log("Doubled:", double);

// 11. filter() function
let ages = [12, 18, 15, 21, 25];

let adults = ages.filter(function(age) {
    return age >= 18;
});

console.log("Adults:", adults);

// 12. reduce() function
```

```
let values = [10, 20, 30, 40];

let total = values.reduce(function(sum, value) {
    return sum + value;
}, 0);

console.log("Total:", total);
```

5. Compilation / Execution Steps

- Step 1 Create a project folder named lab3_exp3/, and inside it a subfolder lab3_2/.
- Step 2 Create exp3_2.js inside lab3_2/ and paste the JavaScript source code.
- Step 3 Open the lab3_exp3/ folder in Visual Studio Code.
- Step 4 Open the integrated terminal (Ctrl + `) in VS Code.
- Step 5 Navigate to the lab3_2/ folder: cd lab3_2
- Step 6 Run the program using Node.js: node exp3_2.js
- Step 7 Verify the console output shows the result of every function call in sequence.

6. Expected Output

The output below is the actual console result produced by running exp3_2.js with Node.js, showing every function type — no/one/multiple parameters, return values, default parameters, function expressions, arrow functions, and array methods (forEach, map, filter, reduce) — in order.

Terminal Output

```
Welcome to JavaScript
Hello Akshaya
Sum: 30
Square: 25
Name: Akshaya
Branch: AIML
Hello Student
Hello Akshaya
Multiplication: 20
Division: 4
Mark: 80
Mark: 75
Mark: 90
Mark: 85
Doubled: [ 2, 4, 6, 8, 10 ]
Adults: [ 18, 21, 25 ]
Total: 100
```

7. Result / Conclusion

The experiment was successfully completed. A JavaScript program was written and executed to demonstrate functions — declarations with no, one, and multiple parameters, return values, string parameters, default parameters, function expressions, and arrow functions — along with the array iteration methods `forEach()`, `map()`, `filter()`, and `reduce()`. The program was run using Node.js, and the console output confirmed correct behaviour of every function and method used, fulfilling Course Outcome CO1.

Student Signature

Faculty / Lab In-charge Signature